Project Report: New York Oil & Gas Production Performance & Decline Analysis

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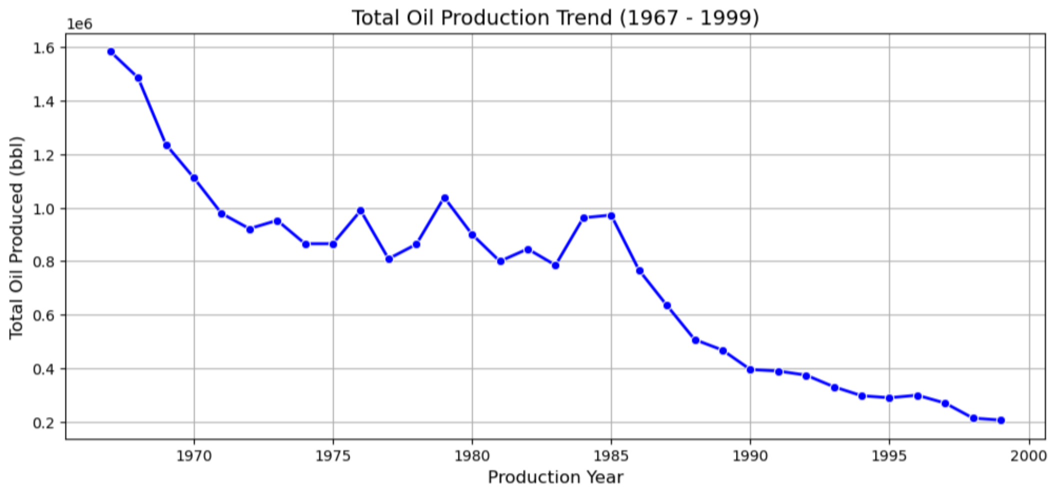
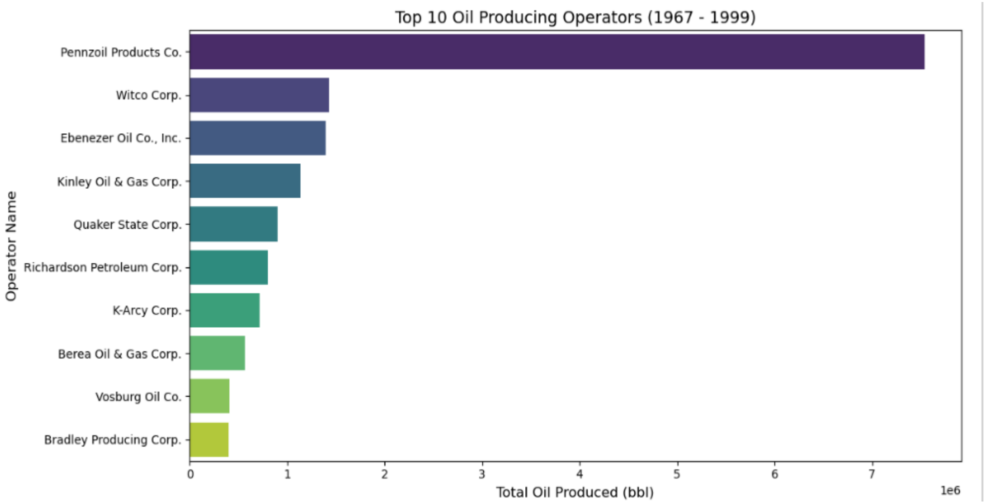
1. Executive Summary

The objective of this project is to analyze the historical oil and gas production performance and decline trends across various counties in New York. Using an end-to-end data analytics approach (Python, SQL, and Tableau), this report identifies the top-producing regions, analyzes multi-year decline patterns, and provides an interactive framework for business stakeholders to make data-driven decisions.

2. Data Source & Preparation (ETL)

- **Dataset Size:** 50,000+ historical production records.
- **Data Cleaning (Python/Pandas):** * Handled missing values and standardized column formats.
  - Filtered data to isolate the core production timeline (1967–1999).
  - Exported the polished dataset as `cleaned_oil_and_gas_data.csv`.





3. Advanced Data Querying (SQL Insights)

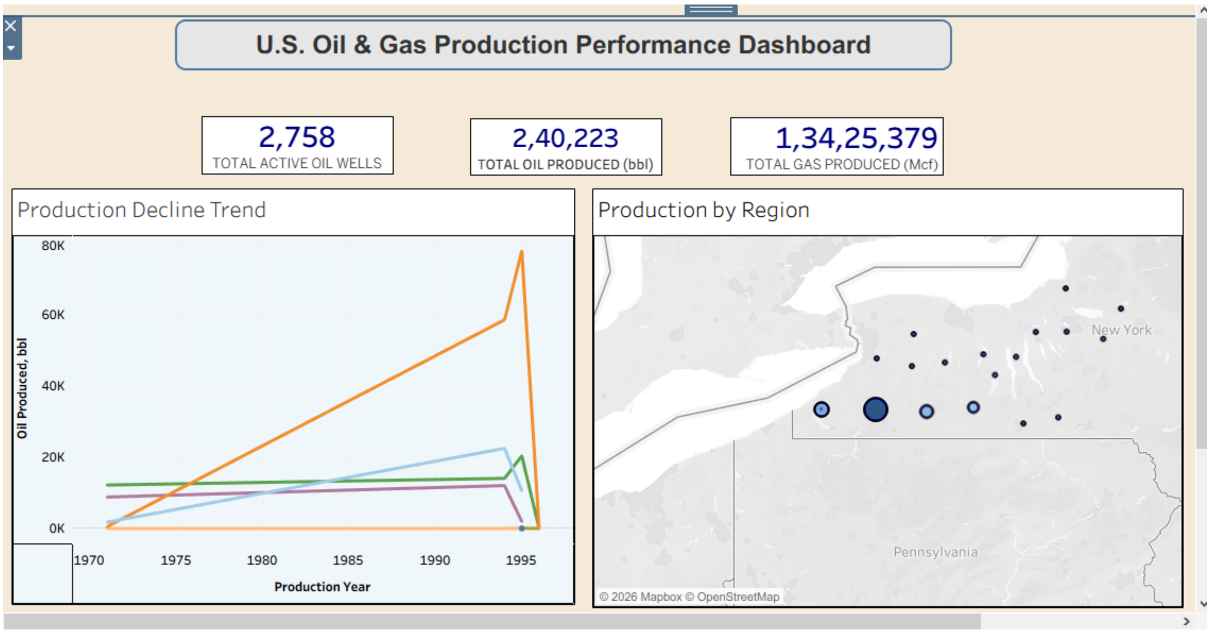
Using MySQL database window functions and aggregation, the following business questions were resolved:

- **Cumulative Production:** Calculated the total volumes extracted across the state.
- **Regional Ranking:** Used `DENSE_RANK()` to isolate the highest-producing counties. **Cattaraugus County** was identified as the top oil producer with a total volume of **1,36,342 bbl**.
- **Operator Performance:** Evaluated production outputs for over 100+ active operators to distinguish market leaders.

4. Data Visualization & Dashboard Design (Tableau)

A centralized executive dashboard, **"U.S. Oil & Gas Production Performance & Analytics,"** was built with the following core components:

- **Summary Cards (KPIs):** Displays high-level metrics including **Total Active Oil Wells (2,758)**, **Total Oil Produced (2,40,223 bbl)**, and **Total Gas Produced (1,34,25,379 Mcf)**.
- **Geospatial Map:** A regional map displaying county-wise production via dynamic bubble sizing.
- **Trend Analysis:** A line chart depicting the production decline curve over the years, clearly highlighting the production peak and subsequent drop-off around 1995.
- **Interactive UI:** Configured dashboard cross-filtering (`Use as Filter`), allowing users to click any region on the map to dynamically update the entire dashboard's charts and KPIs instantly.



5. Key Business Recommendations

1. **Targeted Asset Management:** Resources should be heavily focused on Cattaraugus and high-performing regions identified on the geospatial map, as they drive the majority of the state's cumulative volume.
2. **Decline Interventions:** The sharp production drop-off post-1995 suggests a need for modern secondary recovery methods or well-rejuvenation strategies if these assets are to be reactivated.
3. **Data Democratization:** The automated interactive dashboard should be leveraged by stakeholders for rapid quarterly performance audits without requiring manual data compilation.